

CPU Architecture & Register Organization

Q. Explain the internal organization of a CPU. Describe the basic components including Registers, ALU, and the Control Unit.

> **Definition to Remember**

1. Registers

High-speed, temporary storage locations situated directly on the CPU chip.

2. Arithmetic and Logic Unit (ALU)

The execution core where actual data processing happens.

3. Control Unit (CU)

The manager of the CPU. It does not process data itself.

4. Internal CPU Organization

The Registers, ALU, and CU are connected via an **Internal CPU Bus**. This localized data pathway ensures high-speed data transfer between CPU components without having to use the slower external system bus.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**